

PICKLE LAKE, ON, Canada

WMO: 718450

Lat: 51.4492N

Lon: 90.2181W

Elev: 1267

StdP: 14.04

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-24.1	-35.3	0.8	-28.0	-31.0	1.0	-23.7	19.0	7.8	17.7	6.7	6.6	280	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	83.4	65.8	79.9	63.3	77.0	62.0	69.2	78.7	66.8	74.8	64.7	72.6	10.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	100.6	72.9	63.9	93.1	71.0	61.7	86.2	68.9	34.2	79.4	32.2	74.3	30.5	72.5	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.4	17.3	15.5	-34.2	90.1	5.1	4.1	-37.9	93.0	-40.8	95.4	-43.7	97.8	-47.4	100.8		
			-34.4	73.0	5.1	3.2	-38.0	75.3	-41.0	77.1	-43.8	78.9	-47.5	81.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	-1.8	3.4	17.5	32.6	47.1	59.5	64.5	62.2	52.3	37.6	20.6	4.7
	DBStd	25.56	13.54	13.05	13.68	11.29	9.97	8.20	6.50	6.75	8.64	8.57	12.22	14.15
	HDD50	7403	1606	1306	1009	527	179	16	1	2	74	397	881	1405
	HDD65	11661	2071	1726	1473	972	558	200	87	136	389	848	1331	1870
	CDD50	1385	0	0	1	5	90	302	451	380	143	13	0	0
	CDD65	169	0	0	0	0	3	36	72	49	9	0	0	0
	CDH74	1393	0	0	0	2	51	348	571	362	58	1	0	0
	CDH80	305	0	0	0	0	8	80	134	76	7	0	0	0
	WSAvg	7.9	7.5	7.5	7.8	7.9	8.3	7.9	7.7	8.0	8.4	8.7	8.3	7.4
	PrecAvg	28.9	1.3	1.1	1.4	1.6	2.8	3.7	4.1	3.5	3.4	2.5	2.1	1.5
(15) Precipitation	PrecMax	36.3	2.3	1.8	3.0	4.3	4.6	7.4	6.5	6.3	8.0	4.0	4.5	2.8
	PrecMin	21.0	0.6	0.5	0.5	0.4	0.8	1.1	1.3	1.1	0.4	0.9	0.9	0.6
	PrecStd	4.5	0.4	0.4	0.6	0.9	1.1	1.8	1.4	1.5	1.8	1.1	0.9	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	38.9	54.5	66.9	79.2	87.3	89.5	86.6	79.8	67.3	50.9	36.1
		MCWB	32.0	35.6	45.3	50.3	58.8	67.0	70.3	67.8	64.3	56.6	45.5	33.0
	2%	DB	29.0	33.3	46.7	60.8	74.0	81.5	83.4	81.6	74.9	61.1	44.5	32.0
		MCWB	28.1	30.6	39.1	46.3	55.7	62.8	66.3	66.2	62.6	52.9	40.3	30.8
	5%	DB	24.9	29.4	41.5	54.8	69.6	78.0	79.8	78.0	70.3	55.7	40.0	28.7
		MCWB	23.8	26.8	35.4	42.3	54.0	60.8	63.5	64.2	59.8	49.5	37.3	27.6
	10%	DB	19.2	23.9	36.6	49.9	64.8	74.2	76.7	74.4	65.9	50.6	36.2	25.0
		MCWB	18.3	21.9	32.0	39.4	51.9	58.9	62.3	62.3	57.8	46.6	34.2	24.1
	0.4%	WB	32.2	36.2	46.1	52.1	63.8	70.9	72.3	71.5	67.9	58.8	46.9	33.6
		MCDB	32.9	38.3	50.9	66.1	72.2	82.5	85.3	81.4	74.9	65.5	50.0	34.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.2	31.2	39.9	47.1	59.1	66.5	69.3	68.8	64.5	54.0	41.4	30.7
		MCDB	29.0	33.1	45.3	57.9	68.2	75.6	78.6	77.8	71.2	59.5	43.7	32.0
	5%	WB	23.9	27.0	35.8	43.3	55.6	63.7	66.8	66.4	61.8	50.2	37.5	27.7
		MCDB	24.9	29.2	40.7	53.0	65.9	72.1	75.0	73.6	68.6	54.6	39.6	28.7
	10%	WB	18.2	22.0	32.5	40.1	52.8	61.2	64.6	64.2	58.8	46.5	34.0	24.1
		MCDB	19.1	24.2	36.0	49.2	63.2	70.6	72.8	72.1	64.6	50.6	36.0	25.0
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	19.3	20.8	20.5	19.9	19.1	17.9	17.4	15.5	11.5	11.4	14.1
		MCDBR	17.2	17.6	21.8	26.7	26.2	24.0	22.2	21.9	21.7	20.0	13.6	13.8
		MCWBR	16.6	15.3	15.1	15.8	12.8	10.6	9.7	10.0	11.9	13.3	10.5	12.9
	5% WB	MCDBR	16.8	15.8	20.3	24.9	22.5	19.4	19.4	18.8	18.3	18.7	12.9	13.6
		MCWBR	16.3	14.3	14.9	16.0	12.6	10.7	10.5	10.4	12.2	13.3	10.6	12.8
(40) Clear-Sky Solar Irradiance	taub		0.241	0.262	0.273	0.314	0.337	0.348	0.373	0.349	0.330	0.288	0.280	0.233
	taud		2.319	2.282	2.365	2.437	2.484	2.479	2.433	2.510	2.557	2.609	2.440	2.373
	Ebn at Noon		254	277	297	295	291	288	278	280	275	270	236	241
	Edh at Noon		19	27	30	32	32	33	34	30	25	20	18	16
(44) All-Sky Solar Radiation	RadAvg		365	761	1187	1611	1638	1728	1707	1421	1002	584	300	226
	RadStd		43	59	107	126	114	149	161	132	115	79	32	36

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page